CO1 - AT1

SHORT ANSWER TEST

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Course name : Fundamentals of data Science.

Course code : DSA0401

1) A collage wants to identify students who may fail in the final exam. how can data science help?

Ans:

=> Data science can analyze students 'Previous marks', attendance, assignment and examination performance.

=> Using Predictive analytics, it can identify students who are at high risk of failing. The collage can provide additional classes, guidance, or learning materials to those students. Thus, Data science helps improve student performance and reduce failure rates.

3) An online shopping website wants to recommend products to customers. which use of data science is applied here?

Ans:

=> A Recommendation System is used to recommend Suitable Products to customers. it analyzes customers' Previous Purchases, searches, ratings, and browsing behavior.

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=> Based on these patterns, the system predicts products that a customer may prefer. This improves customer experience and can also increase sales.

5) A company wants to reduce customer loss. How can data science support this?

Ans:

=> Data Science can analyze customer purchases, complaints, usage patterns, and interactions with the company.

=> Customer churn prediction can identify customers who are likely to stop using the service. The company can provide discounts, better support, or personalized offers to retain them. Thus, Data science helps reduce customer loss and improve customer retention.

7) A Supermarket wants to know which Products sell more during festivals. How can Data Science help?

Ans:

=> Data Science can analyze previous sales records from different festivals and reasons demand Prediction can identify Products that are likely to have higher demand a festival.

=> The supermarket can use this information to maintain sufficient stock of those products.

9) A farmer wants to improve crop yield. Give one use of data science.

Ans:

=> Data Science can analyze soil conditions, rainfall, temperature, irrigation, and previous crop production related data.

=> Crop yield prediction can estimate the expected production under different conditions.

=> farmers can use these predictions to select suitable crops and improve farming practices.

11) A weather app updates temperature every minute. Which facet of data is involved?

Ans:

=> The big data characteristic involved is velocity.

=> velocity refers to the speed at which data is generated, collected and processed.

=> Therefore, the example represents high data velocity.

13) Customer comments on social media are collected for analysis. what type of data is this?

Ans:

=> Customer comments on social media are generally considered Unstructured data. They do not follow a fixed format such as rows and columns in a database.

=> The comments may contain text, emojis, images, videos, and different languages.

=> Data science techniques can be used to analyze this information and customer opinions.

15) A hospital dataset contains missing age values and incorrect phone numbers. which data quality issue is present?

Ans:

=> The data quality issues are completeness and accuracy.

=> missing age values indicate a lack of completness, because some required information is absent.

=> Therefore, the dataset has both completeness and accuracy problems.

17) A company stores large files in hadoop Distributed file System - which part the big data ecosystem is used?

Ans:

=> The component used is HDFS (Hadoop distributed file system).

=> HDFS is a distributed storage system used to store large across multiple computers.

=> It divides data into blocks and distributes them across different nodes in the clusters.

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19) An Organization wants to collect live sensor data from IOT devices. which big data ecosystem component is suitable?

Ans:

=> Apache Kafka is suitable for collecting and streaming real-time sensor data from IOT devices.

=> It can handle large volume of data generated by multiple sensors in real time.

=> Therefore, Kafka is commonly used for real-time IOT data collection and streaming.